

Theorem 20

The quotient $(A^*/\approx, \odot)$ of $(A^*, \cdot)$ under $\approx$ with $\odot$ defined so $[\alpha]_{\approx} \odot [\beta]_{\approx} = [\alpha \cdot \beta]_{\approx}$ is an intervention algebra.

- $\alpha, \beta \in A^*$ - finite sequence of elements of A

- $[\alpha]_{\approx} \in A^*/\approx$ - the class of α . The quotient elements are classes (buckets) $\rightarrow \odot$ must be an operation on classes

- $[\alpha]_{\approx} \odot [\beta]_{\approx} = [\alpha \cdot \beta]_{\approx} \rightarrow$ to multiply classes: pick members, concatenate, take the class.

Proof:

① Define signature: variables V are the equivalence classes of A , & for each variable, its value range is the set of members of that class

$$V = A/\approx \text{ \& \; } Val_{[\alpha]_{\approx}} = [\alpha]_{\approx}$$

② candidate map: $i^*(x_1, \dots, x_n) = x_1 \odot \dots \odot x_n$

- read the sequence & compose it as hard interventions on Σ

$$\textcircled{3} i^*(\alpha) = i^*(\text{Collapse}(\alpha)) = i^*(\text{Sort}(\text{Collapse}(\alpha)))$$

• $i^*(\alpha) = i^*(\text{Collapse}(\alpha)) \rightarrow$ running the full intervention list on any causal model M produces the exact same intervened model as running the collapsed list

Example:

$A = \{ x := v : v \in \{0 \dots 9\} \} \cup \{ y := w : w \in \{0 \dots 20\} \}$ $\{ b, b \cdot c \cdot a, a \cdot c, c \cdot a, b \cdot a \cdot c, a \cdot a \cdot c, \dots \}$

A has 31 "atoms", $x \sim y$ iff they name the same variable.

3 atoms:

① $a = 'x := 3'$ ② $b = 'x := 5'$ ③ $c = 'y := 7'$

- A has 2 equivalence classes: the ten X-atoms & the 21 Y-atoms.

- At this stage, A is just a set of strings with an equivalence relation.

A^* : all finite sequences. Ex. $\alpha = b \cdot b \cdot c \cdot a$
 $b \cdot b \cdot c \cdot a \xrightarrow{\text{delete}(b \cdot b)} b \cdot c \cdot a \xrightarrow{\text{swap}(c \cdot a)} b \cdot a \cdot c \xrightarrow{\text{delete}(b \cdot a)} a \cdot c$

$\Rightarrow \boxed{\alpha \approx a \cdot c} - *$

- Normal Form: Fix order $<$ on $\sim$ classes:
 $[X\text{-atoms}] < [Y\text{-atoms}]$. Compute:

Collapse(α): scan $b \cdot b \cdot c \cdot a$ & delete every atom with $\sim$ equivalence to its right

First b has a to its right $\rightarrow$ delete. Second b has a to its right $\rightarrow$ delete. Result: $c \cdot a$

Sort(α): X-atom sorts before Y $\rightarrow \boxed{a \cdot c}$

Sort(Collapse(α)) = $a \cdot c$ - matching rewrite *

- The bucket is $[\alpha] \approx$:

$\{ b \cdot b \cdot c \cdot a, a \cdot c, c \cdot a, b \cdot a \cdot c, a \cdot a \cdot c, \dots \}$
 * infinitely many sequences, one shared normal form $a \cdot c$, one element of $A^*/\approx$.

- Manufacture the signature:

variables = $\sim$ -classes & values = members of each variable

$V = \{X, Y\}$, $Val_X = \{ "x := 0", \dots, "x := 9" \}$

$Val_Y = \{ "y := 0", \dots, "y := 20" \}$

- Causal Model: $M = (\Sigma, \{F_X, F_Y\})$

$F_X(v) = 0$ (constant, input variable)

$F_Y(x) = 2x$ ($X \propto Y$, Y depends on X)

Solve(M) = $\{x=0, y=0\}$ with no interven.

• M_a replaces F_X with constant function

$v \mapsto 3$. Solve(M_a): $\{x=3, y=6\}$

• $L^*(\alpha) = b \cdot b \cdot c \cdot a$ applied to M :

original M	F_X $v \mapsto 0$	F_Y $x \mapsto 2x$
b	$v \mapsto 5$	$x \mapsto 2x$
b	$v \mapsto 5$	$x \mapsto 2x$
c	$v \mapsto 5$	$v \mapsto 7$
a	$v \mapsto 3$	$v \mapsto 7$

• Final model: both mechanisms constant
 Solve = $\{x=3, y=7\}$. Now evaluate the
 normal form instead: $L^*(a.c) = a \circ c$
 replaces F_x with $v \mapsto 3$ & F_y with $v \mapsto 7$
 identical to the final model, reached in 2
 steps instead of 4. $L^*(c.a)$: same two
 replacements, different order, same result.

$$L^*(\underbrace{b.b.c.a}_{\alpha}) = L^*(\underbrace{c.a}_{\text{collapse}(\alpha)}) = L^*(\underbrace{a.c}_{\text{sort}(\text{collapse}(\alpha))}) = \{x=3, y=7\} \in \Phi \quad \text{Hard intervention}$$

• $L([\alpha]_{\approx}) = L^*(\alpha) \rightarrow L^*$ is constant across the bucket

- Little exercise: compute $L([b.c]_{\approx} \odot [a]_{\approx})$

① left path: $[b.c] \circ [a] = [b.c.a] \rightarrow \text{collapse } b$
 leaving $c.a$, sort gives $a.c \rightarrow \{x=3, y=7\}$

② right path: $L([b.c]) \circ L(a) = \{x=5, y=7\} \circ \{x=3\}$

if we perform first intervention then second, F_x
 overrides: $\{x=3, y=7\} \rightarrow \text{Same answer!}$

Equality holds for all pairs of buckets!